

# Shubhadeep Mandal

☎ 81602 63797   ✉ shubhadeepmandal11@gmail.com   💻 shubhadeep-mandal   🎧 Seronic2001

## EDUCATION

|                                                                                                                                                                   |                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>International Institute of Information Technology, Hyderabad</b><br><i>Telangana, India</i><br><i>Master of Technology in Computer Science and Engineering</i> | July 2025 – Present<br><br><i>CGPA: 9.12</i> |
| <b>Indian Institute of Information Technology, Ranchi</b><br><i>Ranchi, India</i><br><i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>    | Aug 2020 – Jun 2024<br><br><i>CGPA: 9.05</i> |

## PROJECTS

- AmbiStory: NLP Plausibility & Disambiguation Engine** | *Python, PyTorch, Transformers, NLI* 🔗
- Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching  $\rho = 0.535$  and 72.2% test accuracy.
  - Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head.
  - Boosted generalization by +0.265  $\rho$  across unseen homonyms using group-stratified 5-fold CV, annotator KL divergence loss, and weight averaging.
- Football Video Analytics with YOLO** | *Python, YOLOv8, OpenCV, Optical Flow, K-Means* 🔗
- Fine-tuned YOLOv8 for multi-object tracking of players, ball, and referees, with K-Means color clustering for team assignment.
  - Compensated camera pan/tilt using Lucas-Kanade optical flow and mapped pixel coordinates to pitch meters via homography.
  - Engineered continuous speed, distance covered, and team possession telemetry pipelines with real-time video overlay generation.
- WatchTower AI: Real-Time Edge Object Detection** | *Next.js 14, TypeScript, TensorFlow.js, WebRTC* 🔗
- Built an in-browser edge object detection web application using TensorFlow.js and quantized COCO-SSD, achieving 30 FPS zero-latency inference.
  - Processed live camera feeds via WebRTC APIs, rendering dynamic bounding boxes, confidence scoring, and snapshot recording on HTML5 Canvas.
- Peer-to-Peer Distributed File Sharing System** | *C++, Sockets, Multithreading, OpenSSL* 🔗
- Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers.
  - Engineered out-of-order parallel chunk writing via `pwrite()` and dynamic bitfield generation for simultaneous partial seeding.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, TypeScript, JavaScript, SQL, Bash.  
**NLP & Deep Learning:** Transformers, PyTorch, Hugging Face, Sentence-Transformers, NLI, Ordinal Regression.  
**Computer Vision & Edge ML:** YOLOv8, OpenCV, TensorFlow.js, Optical Flow, K-Means, Streamlit.  
**Systems, Data & Tools:** POSIX Sockets, Linux/Unix, Git, NumPy, Pandas, Scikit-learn, Matplotlib.

## ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

## INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.